

ALEKSEI PETRENKO

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Reinforcement learning expert: LLM post-training/RLVR, robotics, autonomous driving. Versatile researcher and engineer with deep expertise in C++/Python, software architecture, simulation, high-throughput systems.

EDUCATION

PhD in Computer Science at University of Southern California 2018–2022, Los Angeles, USA

Advisor: Prof. [Gaurav Sukhatme](#), [USC Robotic Embedded Systems Lab](#).

Thesis: High-Throughput Methods for Simulation and Deep Reinforcement Learning.

Awarded the USC Provost PhD Fellowship.

Research areas: deep RL, machine learning systems, high-throughput simulation, population-based training, multi-agent envs and self-play, robotics, sim-to-real.

M.Sc. & B.Sc. in Computer Science at Nizhny Novgorod State Technical University 2008–2014, Russia

Both summa cum laude: M.Sc. GPA 5.0/5.0; B.Sc. GPA 4.95/5.0.

Scholarship of the President (1 of 2 recipients in a cohort of ~350; top 0.6%).

Represented the university at ACM ICPC programming contests; World Quarter- and Semifinals, NEERC.

EXPERIENCE

15 years in industry, from hardcore software engineering to cutting-edge AI research.

Sr. Research Scientist at Apple 2023–present, Cupertino, USA (hybrid)

Reinforcement learning research at scale in [Vladlen Koltun](#)'s group. Built verifiable training tasks, environments, harnesses. Owned the internal RL training framework and the key training runs across multiple projects:

- “Population-Based Post-Training” (under review): originated and co-led the project. Asynchronous PBT for LLM post-training that automatically tunes hyperparameters; beats standard RLVR on four agentic benchmarks.
- “[Entropy-Preserving RL](#)”: originated and owned the project. Studied how policy gradient updates and numerical artifacts affect policy entropy; introduced entropy-preserving methods REPO and ADAPO; set a new SOTA on the agentic benchmark AppWorld.
- “[LOOP](#)”: RL for long-horizon, multi-turn, tool-using LLM agents. A 32B Qwen2.5 LoRA fine-tune reached 71% on AppWorld, +9 points over OpenAI o1 despite training on trajectories only from 24 scenarios.

Earlier: co-first author of “[GIGAFLow](#)” (ICML 2025), self-play RL for autonomous driving at 1 trillion sim. transitions (1.6B km). A single policy, trained without human data, is zero-shot SOTA on CARLA, nuPlan, and Waymax.

Technologies: Python, PyTorch, FSDP2, vLLM, WandB, Claude Code, [autoresearch](#), Ray, multi-node clusters.

PhD Researcher, Robotics and Dexterity at [NVIDIA](#) 2021–2022, Los Angeles, USA (hybrid)

- Created GPU-accelerated vectorized robotic simulations featuring automatic domain randomization (ADR) for zero-shot sim-to-real transfer.
- Developed large-scale RL and population-based training algorithms to learn in-hand manipulation ([video](#)), as well as object grasping and placement with hand-arm systems ([video](#)).
- This work resulted in two publications: “[DeXtreme](#)” (in-hand manipulation with a real robotic hand, ICRA 2023) and “[DexPBT](#)” (dexterous control of high-DoF hand-arm systems, RSS 2023, first author).

Technologies: Python, PyTorch, PhysX, Isaac Gym, Docker, WandB.

PhD Researcher, Intelligent Systems at [Intel Labs](#) 2019–2020, Santa Clara, USA

- Created a high-throughput simulator “[Megaverse](#)” with a vectorized renderer capable of rasterizing 1,000,000 agent observations per second on a single node (ICML 2021).

- Developed a new asynchronous RL system that trains at 150,000 FPS on a single GPU in immersive environments with visual observations (“[Sample Factory](#)”, ICML 2020). Using this framework, achieved a new SOTA in VizDoom (agents playing 3D games from pixels), including challenging multi-agent and self-play scenarios.

Technologies: C++17, Python, PyTorch, TensorFlow, RLib, Vulkan, OpenGL, OpenAI Gym, Slurm.

Tech Lead at [itSeez3D](#)

2015–2018, N. Novgorod, Russia

Sr. Software Engineer at [itSeez3D](#)

2013–2015, N. Novgorod, Russia

ItSeez/itSeez3D was the company behind the computer vision library OpenCV, where I worked on computer vision, 3D scanning, geometric deep learning, and virtual reality applications.

- Developed backend (algorithms & server-side applications) and frontend for the first portable consumer [3D scanner](#).
- Led algorithms and systems development for [Avatar SDK](#), an AI-based human digitization engine powered by geometric deep learning.
- Led R&D for the automatic generator of full-body characters (skeleton animation, rendering) and demoed it at major conferences and exhibitions: SIGGRAPH, GDC, and VRX ([example avatar](#) generated from a selfie, 2016).
- Optimized deep neural networks for deployment (quantization, compression, etc.).

Technologies: C++11/14, OpenCV, Python, Django, TensorFlow, Keras, OpenGL, Unity, Unreal, Oculus VR.

Software Engineer at [Tecom](#)

2010–2013, N. Novgorod, Russia

Developed a large, frame-accurate, massively parallel broadcast automation system with many levels of redundancy.

Technologies: C++/Qt, C#, Java, JavaScript, Python, Linux.

SELECTED PUBLICATIONS

* equal contribution. Full list on [Scholar](#).

- **Entropy-Preserving Reinforcement Learning.** [A Petrenko](#)*, B Lipkin*, K Chen, E Wijmans, M Cusumano-Towner, R Giryes, P Krähenbühl. In ICLR, 2026. [arXiv](#)
- **Reinforcement Learning for Long-Horizon Interactive LLM Agents.** K Chen*, M Cusumano-Towner*, B Huval*, [A Petrenko](#)*, J Hamburger, V Koltun, P Krähenbühl. [arXiv](#), 2025.
- **Robust Autonomy Emerges from Self-Play.** M Cusumano-Towner*, D Hafner*, A Hertzberg*, B Huval*, [A Petrenko](#)*, E Vinitsky*, E Wijmans*, T Killian, S Bowers, O Sener, P Krähenbühl, V Koltun. In ICML, 2025. [arXiv](#)
- **Proximal Policy Gradient Arborescence for Quality Diversity Reinforcement Learning.** S Batra, B Tjanaka, M Fontaine, [A Petrenko](#), S Nikolaidis, G Sukhatme. In ICLR (spotlight), 2024. [arXiv](#), [GitHub](#)
- **DexPBT: Scaling up Dexterous Manipulation for Hand-Arm Systems with Population Based Training.** [A Petrenko](#), A Allshire, G State, A Handa, V Makoviychuk. In RSS, 2023. [arXiv](#), [Website](#)
- **DeXtreme: Transfer of Agile In-hand Manipulation from Simulation to Reality.** A Handa*, A Allshire*, V Makoviychuk*, [A Petrenko](#)*, R Singh*, J Liu*, D Makoviychuk, K Van Wyk, A Zhurkevich, B Sundaralingam, Y Narang, J-F Lafleche, D Fox, G State. In ICRA, 2023. [arXiv](#), [Website](#)
- **Megaverse: Simulating Embodied Agents at One Million Experiences per Second.** [A Petrenko](#), E Wijmans, B Shacklett, V Koltun. In ICML, 2021. [arXiv](#), [GitHub](#), [Website](#)
- **Decentralized Control of Quadrotor Swarms with End-to-end Deep Reinforcement Learning.** S Batra*, Z Huang*, [A Petrenko](#)*, T Kumar, A Molchanov, G Sukhatme. In CoRL, 2021. [arXiv](#), [GitHub](#), [Website](#)
- **Agents that Listen: High-Throughput Reinforcement Learning with Multiple Sensory Systems.** S Hegde, A Kanervisto, [A Petrenko](#). In IEEE Conference on Games, 2021. [arXiv](#), [GitHub](#), [Website](#)
- **Large Batch Simulation for Deep Reinforcement Learning.** B Shacklett, E Wijmans, [A Petrenko](#), M Savva, D Batra, V Koltun, K Fatahalian. In ICLR, 2021. [arXiv](#), [GitHub](#)
- **Sample Factory: Egocentric 3D Control from Pixels at 100,000 FPS with Asynchronous Reinforcement Learning.** [A Petrenko](#), Z Huang, T Kumar, G Sukhatme, V Koltun. In ICML, 2020. [arXiv](#), [GitHub](#), [Website](#)

SELECTED OPEN-SOURCE PROJECTS

- [Sample Factory](#) (☆ 1K, 📄 150, 📦 120K): at the time of release the fastest open-source PPO implementation.
- [Megaverse](#) (☆ 230): embodied simulation for AI research. One of the fastest 3D engines, rendering agent observations at over 1,000,000 FPS on a single node.
- [Faster-FIFO](#) (☆ 200, 📦 1.3M): 30x faster alternative (at the time) to Python’s built-in multiprocessing.Queue.

- [Signal-Slot](#): Python implementation of the Qt-like signal & slot async programming paradigm for multiprocessing applications.
- [4DVideo](#): volumetric video recorder and player for Intel RealSense and Google Tango, with a fast real-time Delaunay triangulation (modified Guibas–Stolfi).